

Stella-Kiarie / MovieLens-Recommendation-System-

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This repository features a personalized movie recommendation engine built on the MovieLens Small Dataset. By combining Matrix Factorization (SVD) with a Genre-based Content Filter, the system delivers highly accurate Top-5 recommendations while effectively mitigating the "cold-start" challenge for new users and items.

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Stella-Kiarie

Merge pull request #14 from Stella-Kiarie/stella

3b72c04 · 6 minutes ago

Dataset	Initial Files	3 days ago
README.md	Updated Final README.md	11 minutes ago
non_technical presentation.pdf	Final non-technical presentation	9 minutes ago
student.ipynb	final commit	2 hours ago
student.pdf	Notebook in pdf format	26 minutes ago

README

Movie Recommendation System

1. Project Overview

Movie streaming platforms must keep users engaged by recommending content they are likely to enjoy. With thousands of movies available, users often feel overwhelmed, and poor recommendations can lead to reduced watch time, dissatisfaction, and customer churn.

This project builds a personalized movie recommendation system using the MovieLens Small dataset, combining collaborative filtering and content-based filtering to generate Top-5 movie recommendations for users while addressing data sparsity and cold-start challenges.

Business Problem

- Users are overwhelmed by large movie catalogs
- Poor recommendations reduce engagement and retention
- Platforms need personalized, accurate recommendations

Project Objectives

- Predict ratings for unseen movies
- Generate personalized Top-5 movie recommendations
- Evaluate performance using:
 - i. RMSE (rating prediction)
 - ii. Precision@5 and Recall@5 (Top-N recommendations)
- Address cold-start problems using a hybrid recommendation strategy
- Deliver actionable insights for product and analytics stakeholders

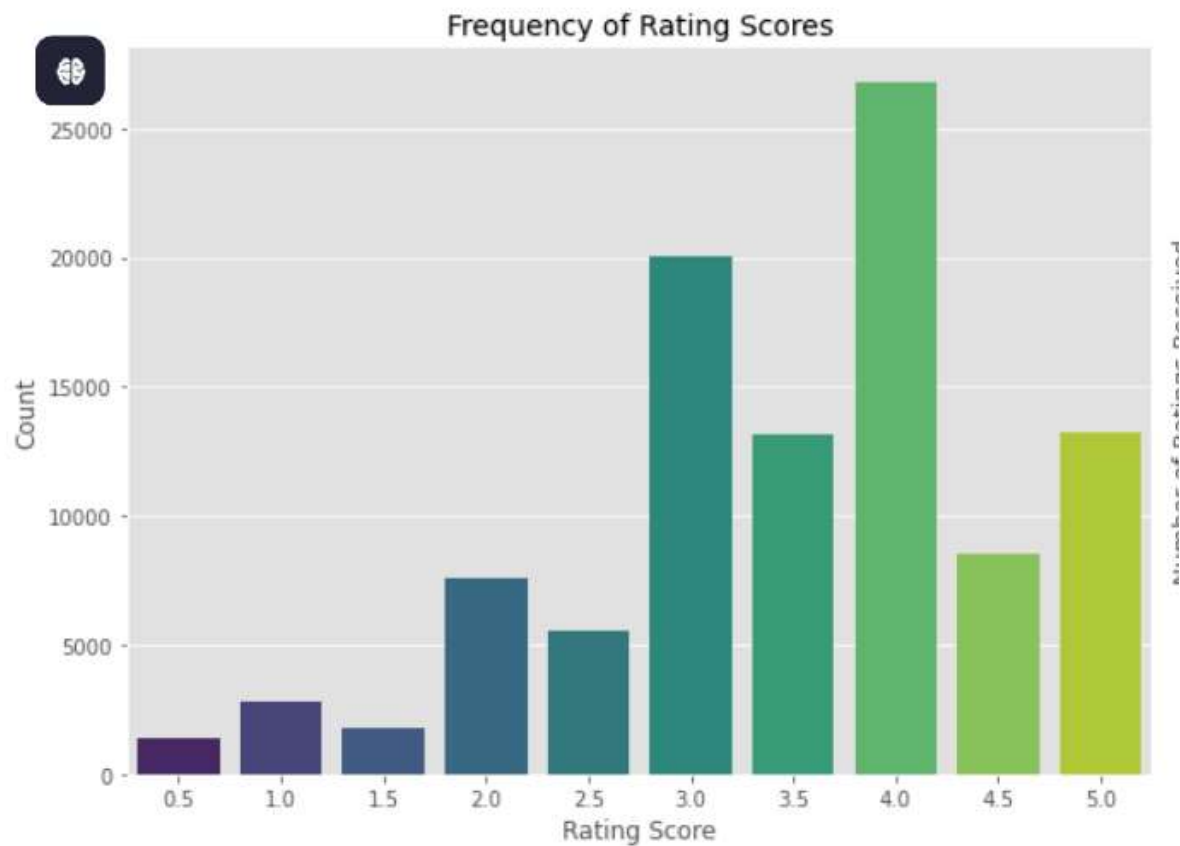
Dataset

Source: MovieLens Small Dataset – GroupLens

- Users: 610
 - Movies: 9,742
 - Ratings: 100,836
 - Rating Scale: 0.5 – 5.0 (explicit feedback)
 - Metadata: Movie titles and genres
- ♦ The dataset is highly sparse (~98.3%), which motivates the use of matrix factorization and hybrid methods.

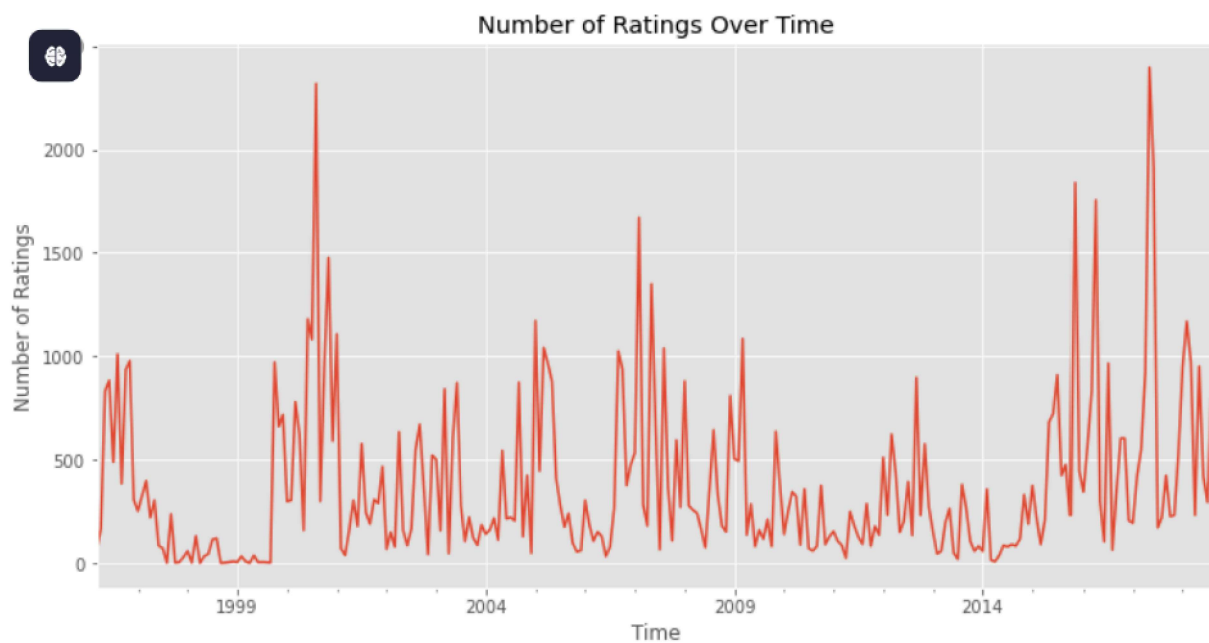
Methodology

1. Exploratory Data Analysis
 - Analyzed rating distributions



Most users rate only a small number of movies, while a few highly active users contribute many ratings.

- Temporal Rating Trends



User activity varies over time, reflecting changes in engagement patterns.

- Identified popularity bias and sparsity patterns
- Explored most-rated and least-rated movies
- Processed genre information for content-based filtering

2. Modeling Approaches

◆ Baseline Model

- Predicts ratings using global and user-level averages
- RMSE: ~0.87

◆ Item-Item Collaborative Filtering (kNN)

- Cosine similarity between movies
- RMSE: ~0.98
- Limited performance due to sparsity

◆ Matrix Factorization (SVD)

- Captures latent user preferences and movie characteristics
- Hyperparameter tuning via GridSearchCV
- Best RMSE: 0.867
- Outperformed baseline and kNN models

Hybrid Recommendation Strategy

To mitigate cold-start issues:

- Primary: SVD-based collaborative filtering
- Fallback: Content-based filtering using movie genres
- Ensures recommendations are available even for:
 - i. New users
 - ii. Newly added movies

This mirrors real-world recommender system architectures.

Evaluation Metrics

Metric Score

1. RMSE (SVD) - 0.867
2. Precision@5 - 0.698
3. Recall@5 - 0.080

Interpretation:

- ~70% of Top-5 recommendations are relevant

- System prioritizes high-quality recommendations
- Lower recall reflects realistic Top-N trade-offs in sparse datasets

Key Insights

- Most ratings fall between 3 and 4
- Users tend to rate movies they already expect to enjoy
- SVD generalizes well in sparse environments
- Hybrid strategy improves robustness and coverage

Tech Stack

- Python
- pandas, numpy – data processing
- matplotlib, seaborn – visualization
- scikit-learn – similarity computations
- Surprise – collaborative filtering models

Results

- SVD with tuning delivers the strongest predictive performance
- Hybrid approach ensures recommendations in cold-start scenarios
- Top-5 recommendations are interpretable, personalized, and relevant
- Visualizations provide actionable insights for business stakeholders

Limitations

- Cold-start issues are reduced but not fully eliminated
- Limited content features (genres only)
- Niche genres may be under-represented

Future Work

- Incorporate richer metadata (actors, directors, keywords)
- Use weighted hybrid models for better balance
- Deploy real-time recommendation pipelines

Releases

No releases published
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Packages

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Contributors 4

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Languages

- Jupyter Notebook 100.0%